Terraform Nginx Docker LaodBalancer

Oleh :

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Proses Instalasi & Pengecekan Package di Ubuntu (WSL)

reshi@DESKTOP-F0HILRR: ~/terra_lab

```
reshi@DESKTOP-F0HILRR:~$ sudo apt install unzip wget -y
[sudo] password for reshi:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-28ubuntu4.1).
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following packages were automatically installed and are no longer required:
  gyp javascript-common libcares2 libfile-basedir-perl libfile-desktopentry-perl libfile-mimeinfo-perl libgles2
  libice6 libio-stringy-perl libipc-system-simple-perl libjs-async libjs-events libjs-inherits libjs-is-typedarray
  libjs-prettify libjs-regenerate libjs-source-map libjs-sprintf-js libjs-typedarray-to-buffer libnet-dbus-perl
  libnotify-bin libnotify4 libre2-10 libsm6 libssl-dev libtie-ixhash-perl libuv1-dev libuv1t64 libx11-protocol-perl
  libxaw7 libxcb-shape0 libxft2 libxkbfile1 libxml-parser-perl libxml-twig-perl libxml-xpathengine-perl libxmu6
  libxt6t64 libxv1 libxxf86dga1 node-abbrev node-ampproject-remapping node-ansi-regex node-ansi-styles node-aproba
  node-are-we-there-yet node-arrify node-async node-async-each node-auto-bind node-babel-plugin-add-module-exports
  node-babel7-runtime node-balanced-match node-base64-js node-binary-extensions node-brace-expansion node-busboy
  node-camelcase node-caniuse-lite node-chownr node-chrome-trace-event node-ci-info node-cjs-module-lexer
  node-cli-boxes node-cli-cursor node-clone node-collection-visit node-color-convert node-color-name node-colors
  node-commander node-commondir node-concat-stream node-console-control-strings node-convert-source-map node-core-js
  node-core-js-pure node-core-util-is node-data-uri-to-buffer node-decompress-response node-deep-is node-defaults
  node-define-property node-delegates node-depd node-diff node-electron-to-chromium node-encoding node-end-of-stream
  node-err-code node-error-ex node-es-module-lexer node-escape-string-regexp node-eslint-utils
  node-eslint-visitor-keys node-esquery node-estraverse node-esutils node-events node-fancy-log node-fast-deep-equal
  node-fast-levenshtein node-fetch node-find-up node-flatted node-foreground-child node-fs-readdir-recursive
  node-fs-write-stream-atomic node-fs.realpath node-function-bind node-get-caller-file node-get-stream node-get-value
  node-glob node-globals node-got node-graceful-fs node-growl node-has-flag node-has-unicode node-has-value
  node-has-values node-hosted-git-info node-iconv-lite node-ieee754 node-iferr node-imurmurhash node-indent-string
  node-inflight node-inherits node-ini node-interpret node-ip node-ip-regex node-is-arrayish node-is-binary-path
  node-is-buffer node-is-descriptor node-is-extglob node-is-path-cwd node-is-plain-obj node-is-plain-object
  node-is-stream node-is-typedarray node-is-windows node-isarray node-isexe node-isobject node-js-tokens
  node-json-buffer node-json-parse-better-errors node-json-schema node-json-schema-traverse node-json-stable-stringify
  node-jsonify node-jsonparse node-kind-of node-levn node-loader-runner node-locate-path node-lodash-packages
  node-lowercase-keys node-lru-cache node-map-visit node-memfs node-merge-stream node-mimic-response node-minimatch
  node-minimist node-minipass node-mute-stream node-n3 node-negotiator node-npm-run-path node-object-inspect
  node-object-visit node-once node-optimist node-optionator node-osenv node-p-cancelable node-p-limit node-p-locate
  node-p-map node-pascalcase node-path-dirname node-path-exists node-path-is-absolute node-path-is-inside
  node-path-type node-pify node-pkg-dir node-postcss-value-parser node-prelude-ls node-process-nextick-args
  node-promise-inflight node-promise-retry node-promzard node-prr node-pump node-punycode node-quick-lru
  node-randombytes node-read node-readable-stream node-rechoir node-regenerator-runtime node-regenerator-transform
  node-regexpp node-regjsgen node-repeat-string node-require-directory node-require-from-string node-resolve
  node-resolve-cwd node-resolve-from node-restore-cursor node-resumer node-retry node-run-queue node-safe-buffer
  node-serialize-javascript node-set-blocking node-set-immediate-shim node-shebang-command node-shebang-regex
  node-shell-quote node-signal-exit node-slash node-slice-ansi node-source-list-map node-source-map node-spdx-correct
  node-spdx-exceptions node-spdx-expression-parse node-spdx-license-ids node-sprintf-js node-ssri node-stack-utils
  node-string-decoder node-strip-bom node-supports-color node-tapable node-terser node-text-table node-through
```

Proses Download, Instalasi Terraform & Docker di Ubuntu (WSL)

reshi@DESKTOP-F0HILRR: ~/terra_lab

```
  node-validate-npm-package-license node-wcwidth.js node-webpack-sources node-wordwrap node-wrappy
  node-write-file-atomic node-xtend node-y18n node-yallist node-yaml x11-utils x11-xserver-utils xdg-utils zutty
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 100 not upgraded.
reshi@DESKTOP-F0HILRR:~$ wget https://releases.hashicorp.com/terraform/1.6.6/terraform_1.6.6_linux_amd64.zip
unzip terraform_1.6.6_linux_amd64.zip--2026-03-02 14:10:42--  https://releases.hashicorp.com/terraform/1.6.6/terraform_1.6.6_linux_amd64.zip
Resolving releases.hashicorp.com (releases.hashicorp.com)... 3.168.203.103, 3.168.203.84, 3.168.203.104, ...
Connecting to releases.hashicorp.com (releases.hashicorp.com)|3.168.203.103|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 24976120 (24M) [application/zip]
Saving to: ‘terraform_1.6.6_linux_amd64.zip’

terraform_1.6.6_linux_amd64.z 100%[===================================================>]  23.82M  4.05MB/s    in 6.2s

2026-03-02 14:10:49 (3.83 MB/s) - ‘terraform_1.6.6_linux_amd64.zip’ saved [24976120/24976120]

reshi@DESKTOP-F0HILRR:~$ unzip terraform_1.6.6_linux_amd64.zip
Archive:  terraform_1.6.6_linux_amd64.zip
  inflating: terraform
reshi@DESKTOP-F0HILRR:~$ sudo mv terraform /usr/local/bin/
reshi@DESKTOP-F0HILRR:~$ terraform -v
Terraform v1.6.6
on linux_amd64
reshi@DESKTOP-F0HILRR:~$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  gyp javascript-common libcares2 libfile-basedir-perl libfile-desktopentry-perl libfile-mimeinfo-perl libgles2
  libice6 libio-stringy-perl libipc-system-simple-perl libjs-async libjs-events libjs-inherits libjs-is-typedarray
  libjs-prettify libjs-regenerate libjs-source-map libjs-sprintf-js libjs-typedarray-to-buffer libnet-dbus-perl
  libnotify-bin libnotify4 libre2-10 libsm6 libssl-dev libtie-ixhash-perl libuv1-dev libuv1t64 libx11-protocol-perl
  libxaw7 libxcb-shape0 libxft2 libxkbfile1 libxml-parser-perl libxml-twig-perl libxml-xpathengine-perl libxmu6
  libxt6t64 libxv1 libxxf86dga1 node-abbrev node-ampproject-remapping node-ansi-regex node-ansi-styles node-aproba
  node-are-we-there-yet node-arrify node-async node-async-each node-auto-bind node-babel-plugin-add-module-exports
  node-babel7-runtime node-balanced-match node-base64-js node-binary-extensions node-brace-expansion node-busboy
  node-camelcase node-caniuse-lite node-chownr node-chrome-trace-event node-ci-info node-cjs-module-lexer
  node-cli-boxes node-cli-cursor node-clone node-collection-visit node-color-convert node-color-name node-colors
  node-commander node-commondir node-concat-stream node-console-control-strings node-convert-source-map node-core-js
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  node-define-property node-delegates node-depd node-diff node-electron-to-chromium node-encoding node-end-of-stream
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  node-eslint-visitor-keys node-esquery node-estraverse node-esutils node-events node-fancy-log node-fast-deep-equal
  node-fast-levenshtein node-fetch node-find-up node-flatted node-foreground-child node-fs-readdir-recursive
  node-fs-write-stream-atomic node-fs.realpath node-function-bind node-get-caller-file node-get-stream node-get-value
  node-glob node-globals node-got node-graceful-fs node-growl node-has-flag node-has-unicode node-has-value
```

reshi@DESKTOP-F0HILRR: ~/terra_lab

```
Setting up containerd (1.7.28-0ubuntu1~24.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1~24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
reshi@DESKTOP-F0HILRR:~$ sudo usermod -aG docker $USER
reshi@DESKTOP-F0HILRR:~$
reshi@DESKTOP-F0HILRR:~$ ls
terraform_1.6.6_linux_amd64.zip  web_laravel
reshi@DESKTOP-F0HILRR:~$ mkdir terra_lab
reshi@DESKTOP-F0HILRR:~$ cd terra_lab
reshi@DESKTOP-F0HILRR:~/terra_lab$ code .
reshi@DESKTOP-F0HILRR:~/terra_lab$ terraform init

Initializing the backend...
Initializing modules...
- backend in modules/backend
- loadbalancer in modules/loadbalancer

Initializing provider plugins...
- Finding kreuzwerker/docker versions matching "~> 3.0"...
- Installing kreuzwerker/docker v3.6.2...
- Installed kreuzwerker/docker v3.6.2 (self-signed, key ID BD080C4571C6104C)

Partner and community providers are signed by their developers.
If you'd like to know more about provider signing, you can read about it here:
https://www.terraform.io/docs/cli/plugins/signing.html

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

reshi@DESKTOP-F0HILRR: ~/terra_lab

```
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
reshi@DESKTOP-F0HILRR:~/terra_lab$
reshi@DESKTOP-F0HILRR:~/terra_lab$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # docker_network.app_network will be created
  + resource "docker_network" "app_network" {
      + driver      = (known after apply)
      + id          = (known after apply)
      + internal    = (known after apply)
      + ipam_driver = "default"
      + name        = "mini_network"
      + options     = (known after apply)
      + scope       = (known after apply)
    }

  # module.backend.docker_container.backend[0] will be created
  + resource "docker_container" "backend" {
      + attach                                      = false
      + bridge                                      = (known after apply)
      + command                                     = [
          + "sh",
          + "-c",
          + "echo '<h1>Backend Server 0</h1>' > /usr/share/nginx/html/index.html && nginx -g 'daemon off;'",
        ]
      + container_logs                              = (known after apply)
      + container_read_refresh_timeout_milliseconds = 15000
      + entrypoint                                  = (known after apply)
      + env                                         = (known after apply)
      + exit_code                                   = (known after apply)
      + hostname                                    = (known after apply)
      + id                                          = (known after apply)
      + image                                       = (known after apply)
      + init                                        = (known after apply)
```

reshi@DESKTOP-F0HILRR: ~/terra_lab

```
Plan: 9 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + backend_container_names = [
      + "backend_0",
      + "backend_1",
      + "backend_2",
      + "backend_3",
      + "backend_4",
    ]
  + load_balancer_url       = "http://localhost:8080"

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Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
reshi@DESKTOP-F0HILRR:~/terra_lab$ terraform applay
Terraform has no command named "applay". Did you mean "apply"?

To see all of Terraform's top-level commands, run:
  terraform -help

reshi@DESKTOP-F0HILRR:~/terra_lab$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # docker_network.app_network will be created
  + resource "docker_network" "app_network" {
      + driver      = (known after apply)
      + id          = (known after apply)
      + internal    = (known after apply)
      + ipam_driver = "default"
      + name        = "mini_network"
      + options     = (known after apply)
      + scope       = (known after apply)
    }

  # module.backend.docker_container.backend[0] will be created
  + resource "docker_container" "backend" {
      + attach                                    = false
      + bridge                                    = (known after apply)
      + command                                   = [
          + "sh",
```
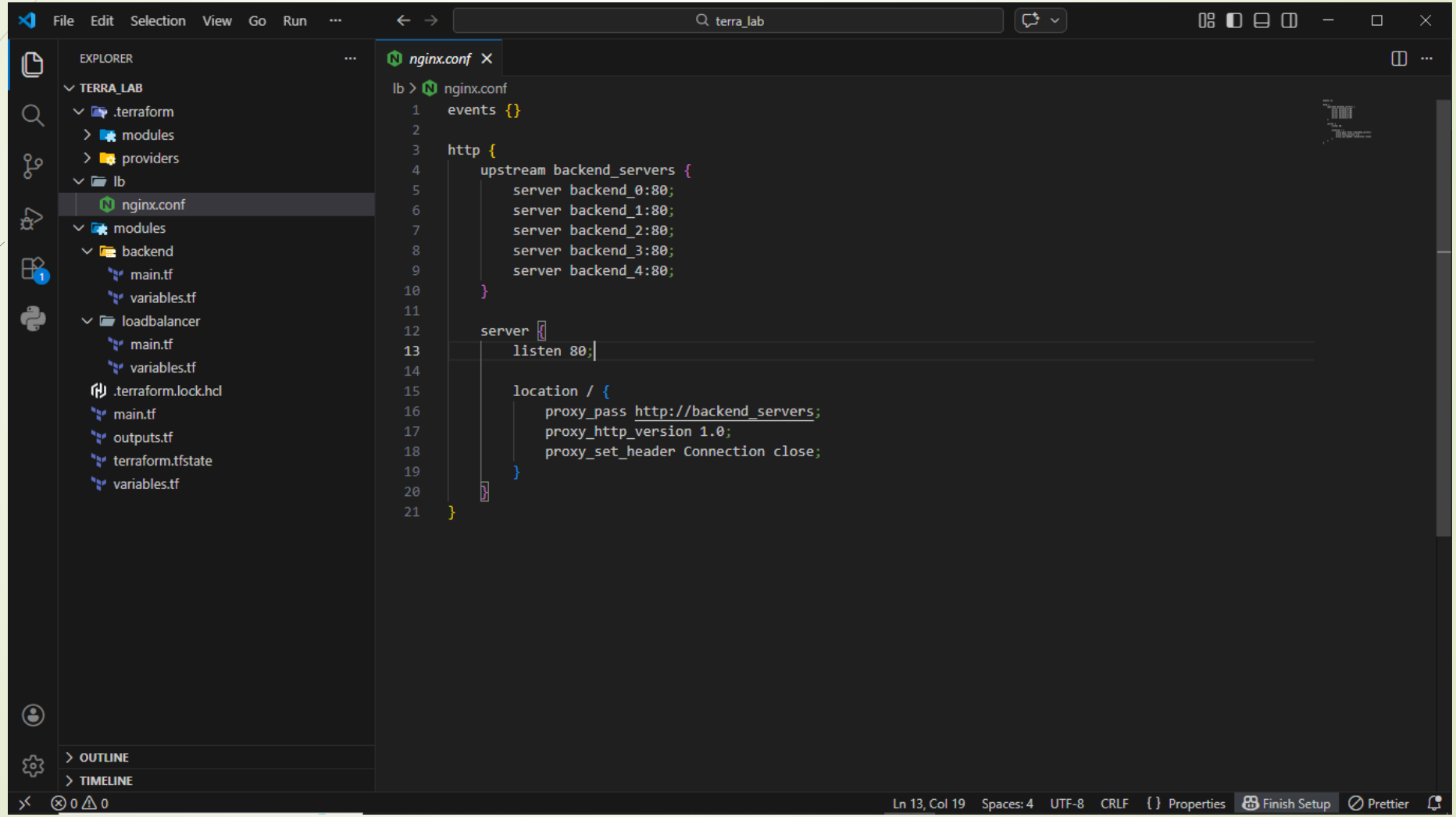
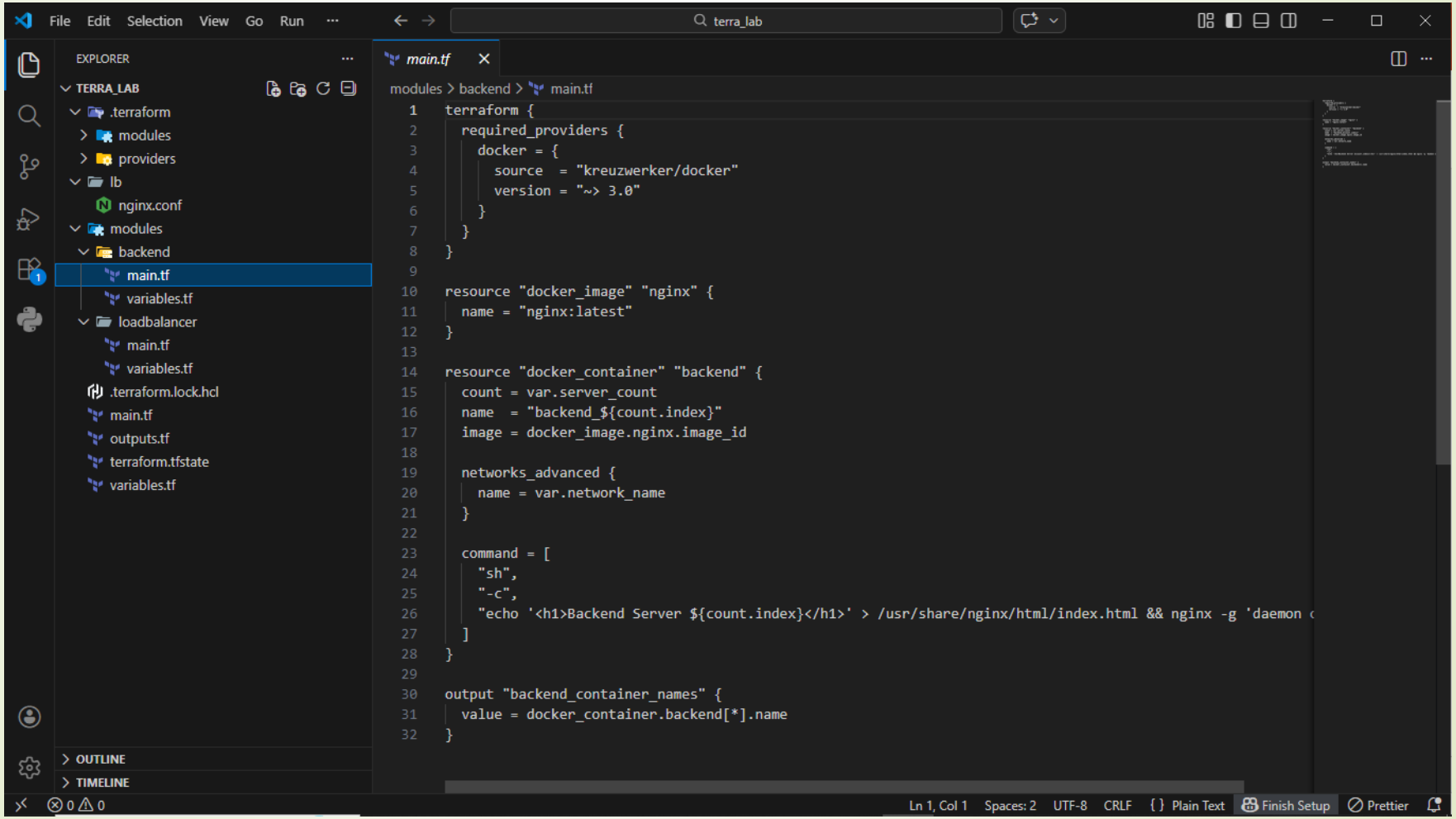
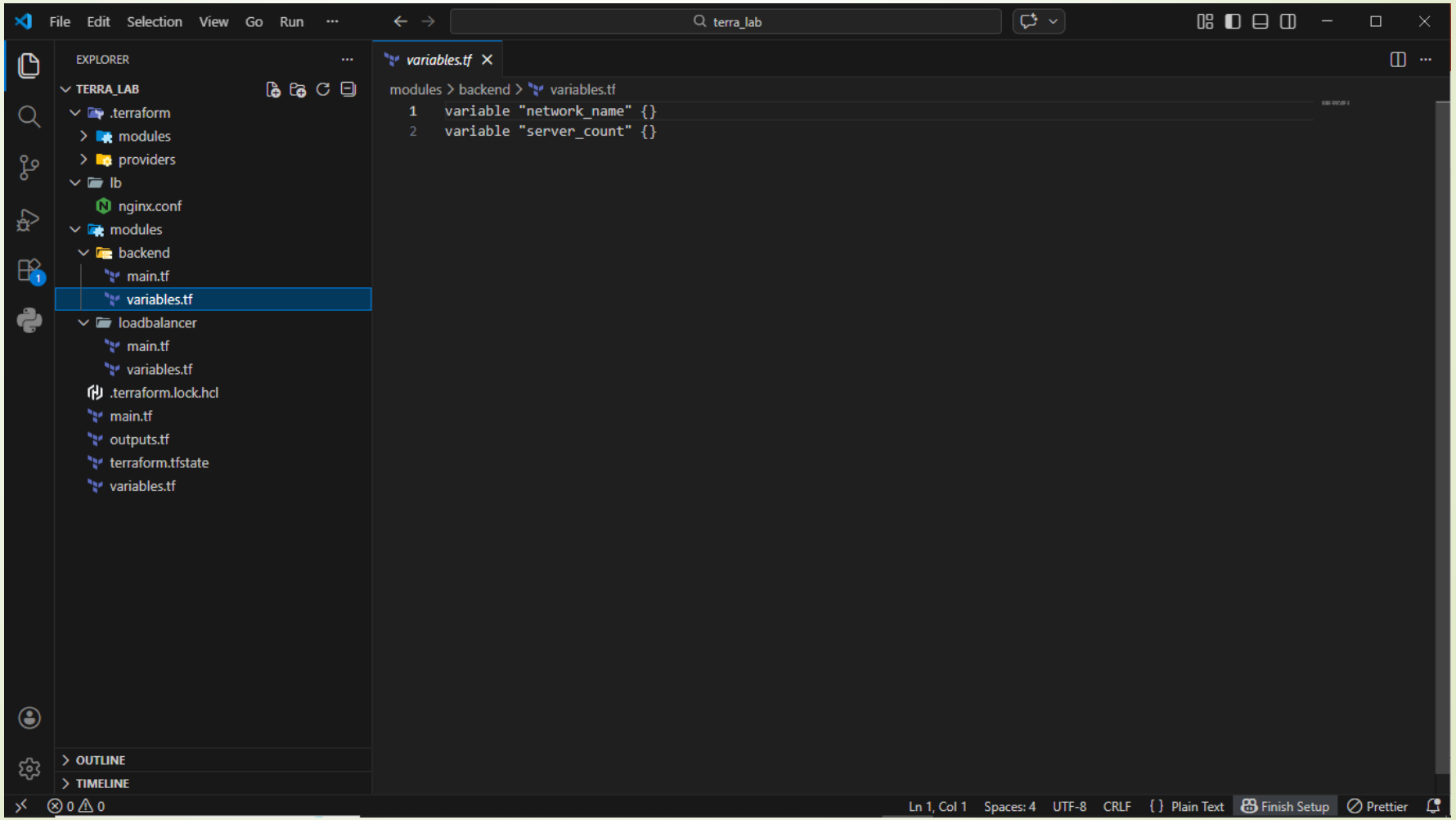

Lib Terra Lab
nginx.conf



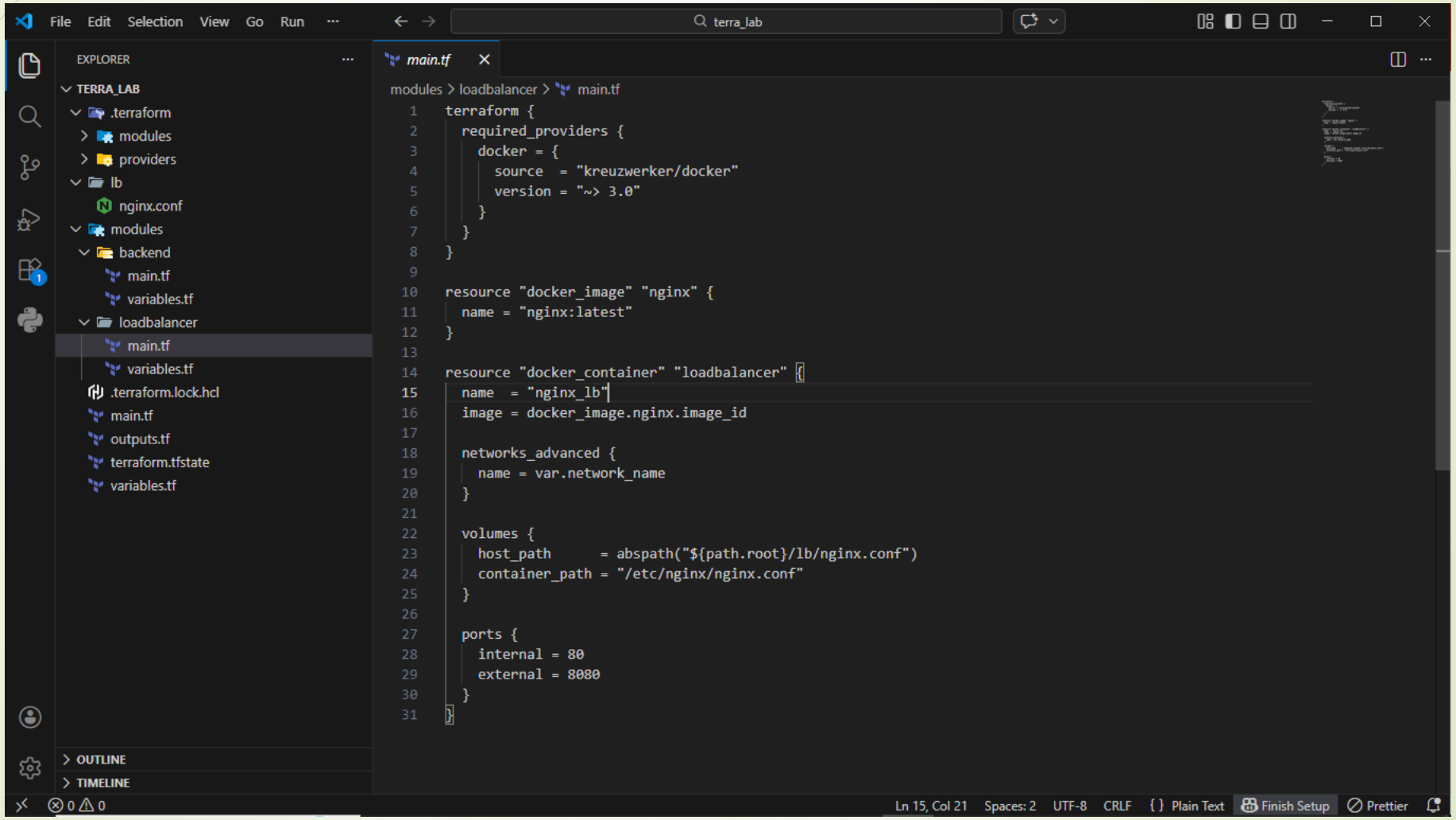
Modules
Backend, main.tf



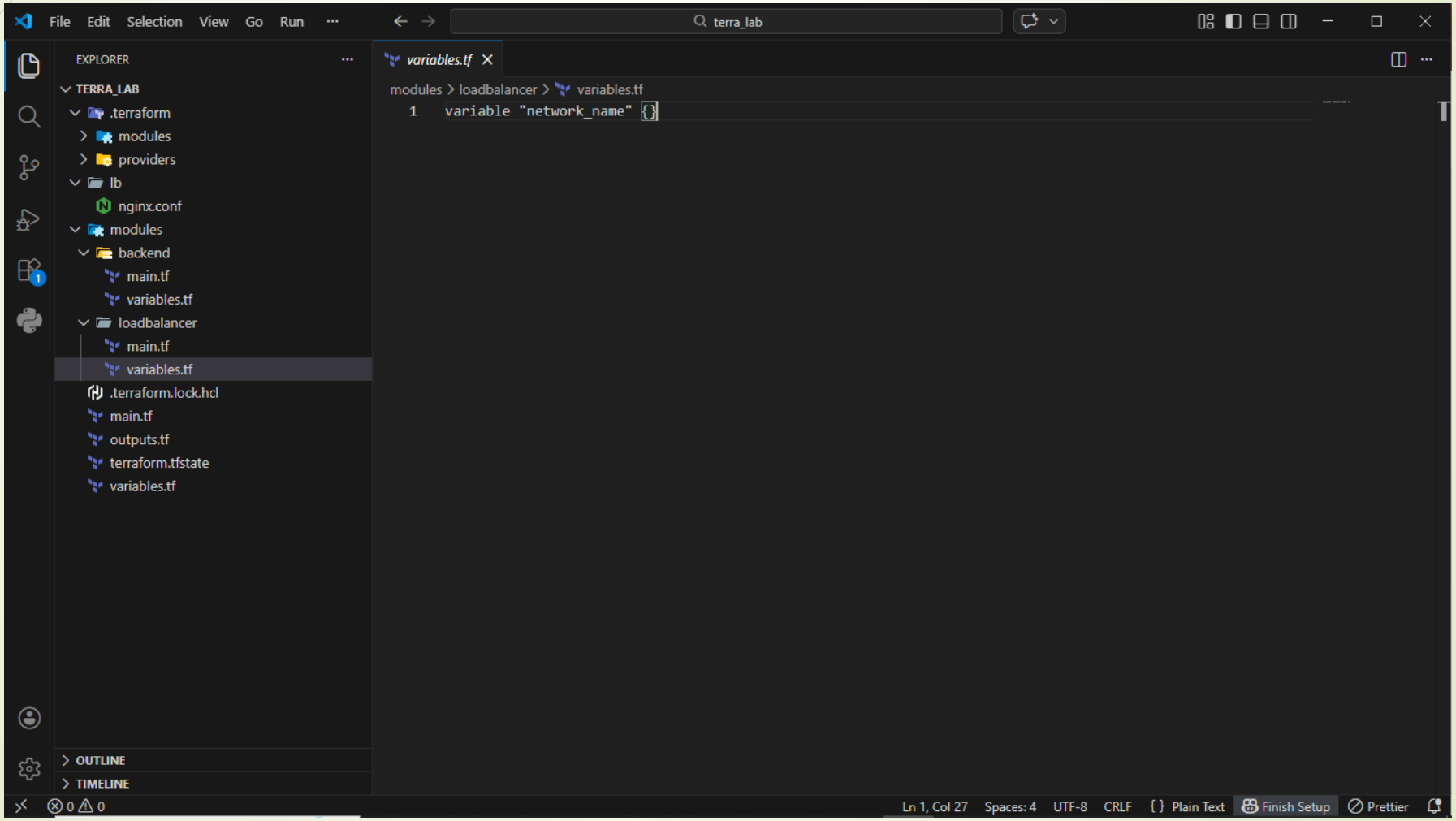
Modules
Backend, variable.tf



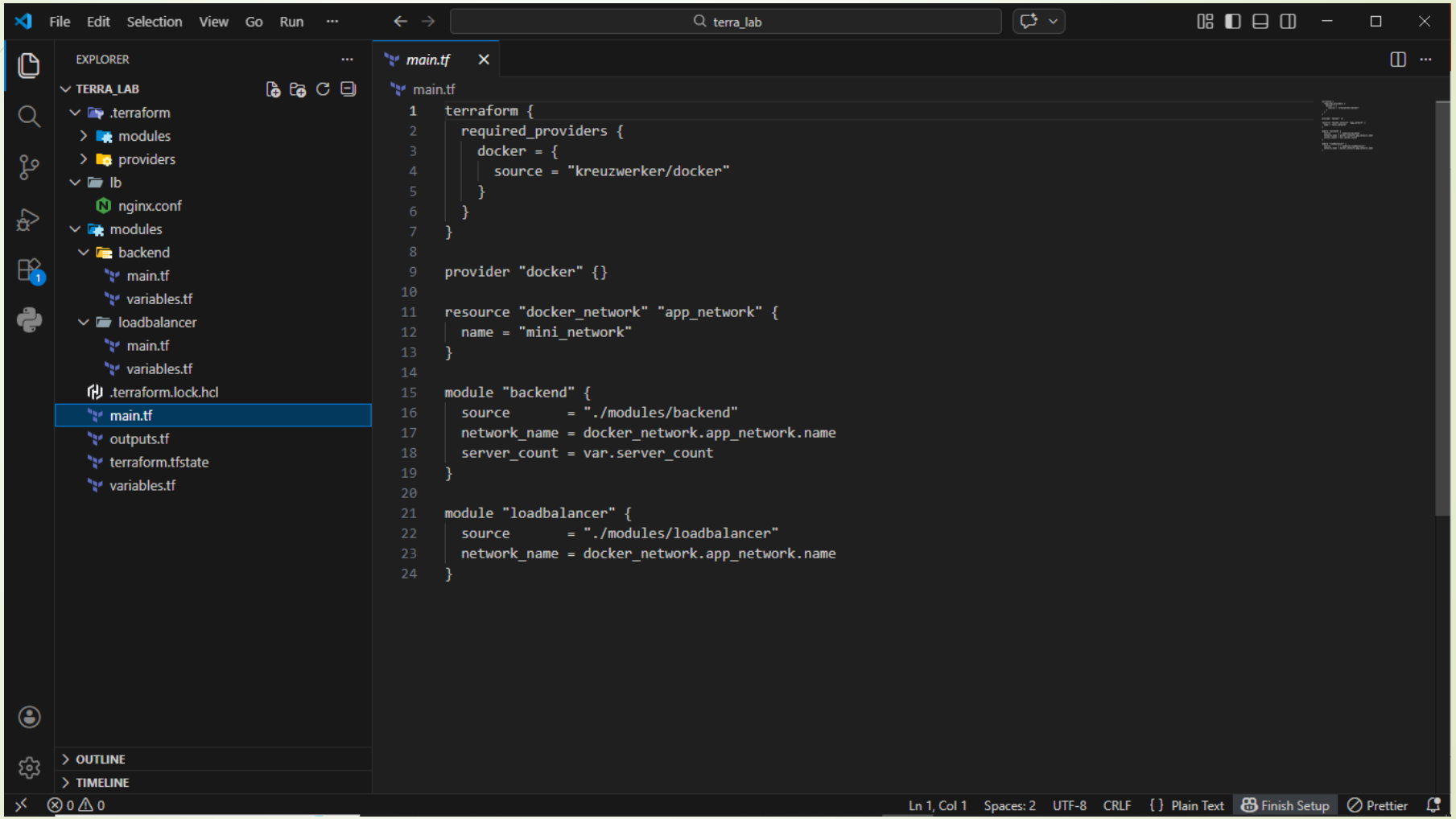
Modules Loadbalancer,main.tf



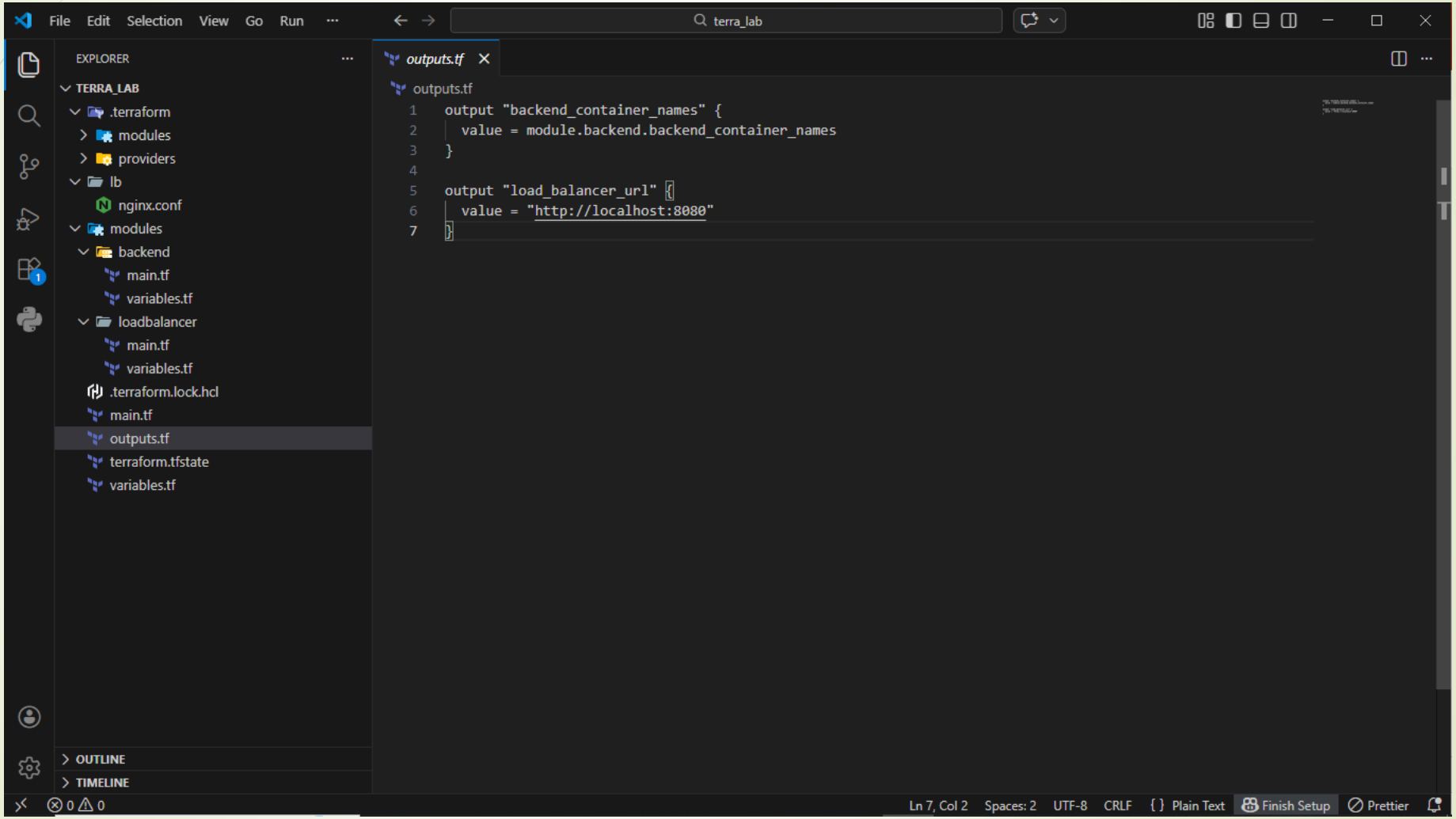
Modules
LoadBalencer, variable.tf



Main.tf


File Edit Selection View Go Run
terra_lab
EXPLORER
TERRA_LAB
.terraform
modules
providers
lb
nginx.conf
modules
backend
main.tf
variables.tf
loadbalancer
main.tf
variables.tf
.terraform.lock.hcl
main.tf
outputs.tf
terraform.tfstate
variables.tf
OUTLINE
TIMELINE
main.tf
main.tf
terraform {
required_providers {
docker = {
source = "kreuzwerker/docker"
}
}
}
provider "docker" {}
resource "docker_network" "app_network" {
name = "mini_network"
}
module "backend" {
source = "./modules/backend"
network_name = docker_network.app_network.name
server_count = var.server_count
}
module "loadbalancer" {
source = "./modules/loadbalancer"
network_name = docker_network.app_network.name
}
0 0
Ln 1, Col 1
Spaces: 2
UTF-8
CRLF
Plain Text
Finish Setup
Prettier

Outputs.tf


File Edit Selection View Go Run
terra_lab
EXPLORER
TERRA_LAB
.terraform
modules
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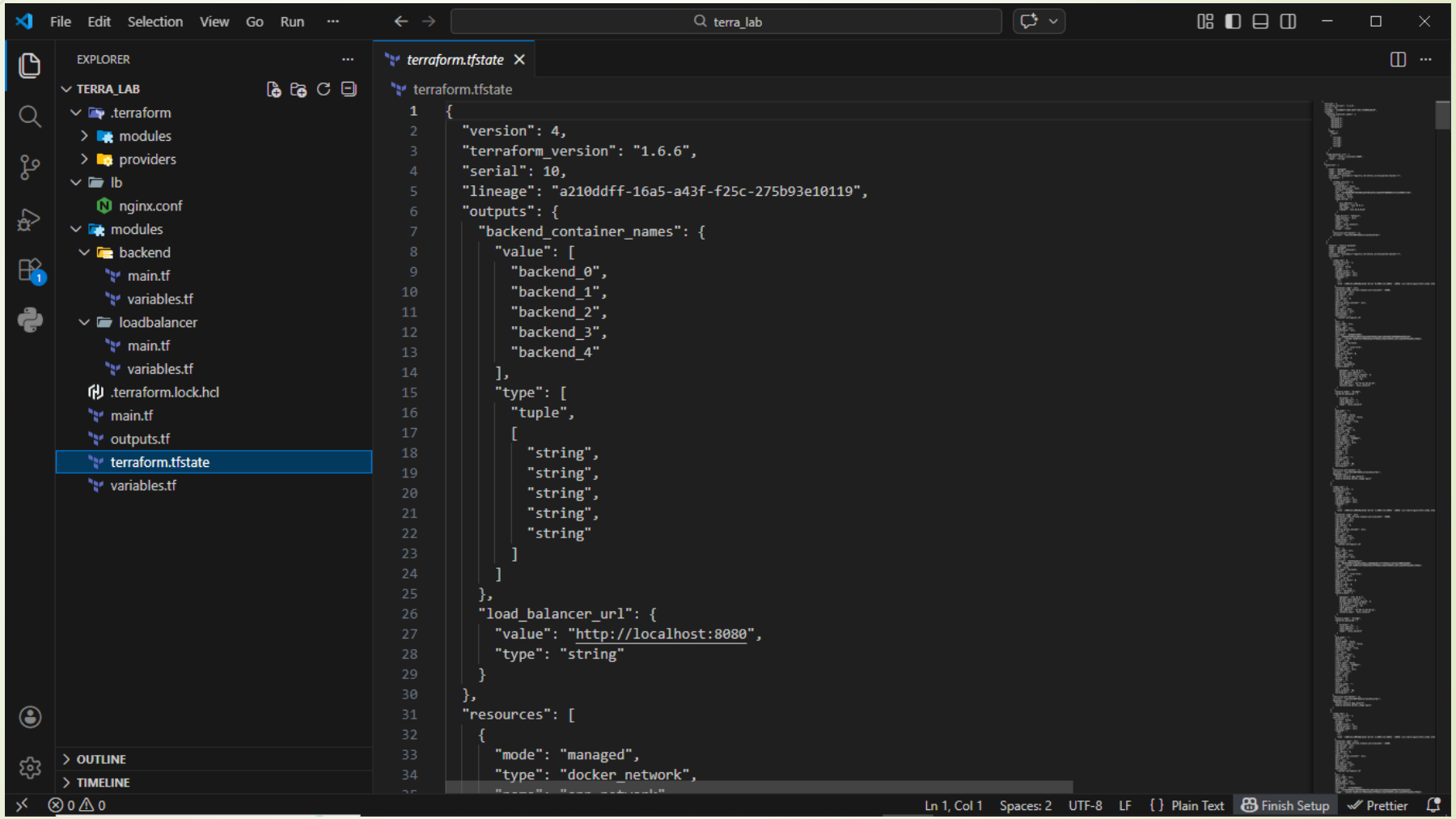
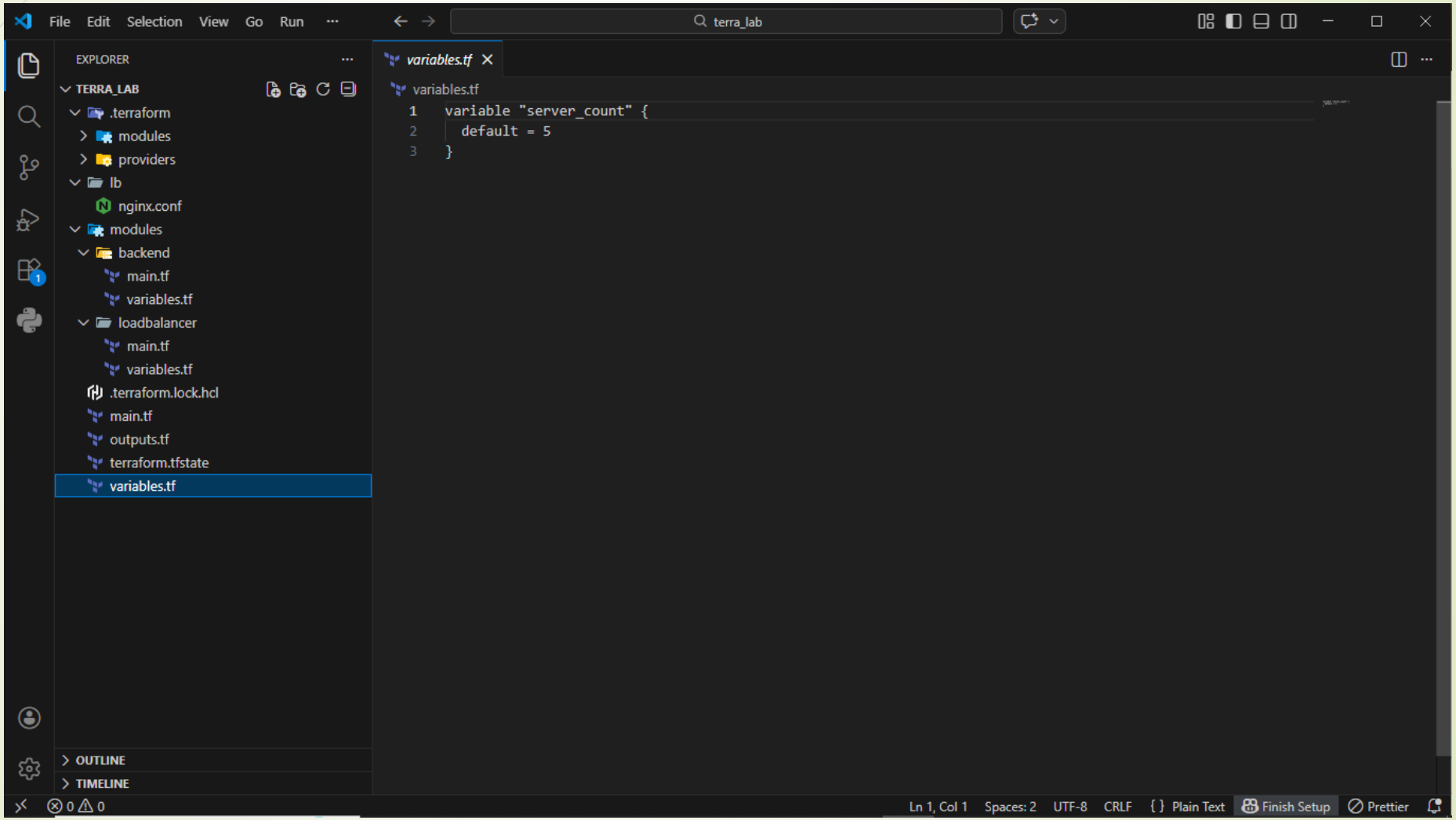
lb
nginx.conf
modules
backend
main.tf
variables.tf
loadbalancer
main.tf
variables.tf
.terraform.lock.hcl
main.tf
outputs.tf
terraform.tfstate
variables.tf
OUTLINE
TIMELINE
outputs.tf
outputs.tf
output "backend_container_names" {
value = module.backend.backend_container_names
}
output "load_balancer_url" {
value = "http://localhost:8080"
}
Ln 7, Col 2
Spaces: 2
UTF-8
CRLF
Plain Text
Finish Setup
Prettier

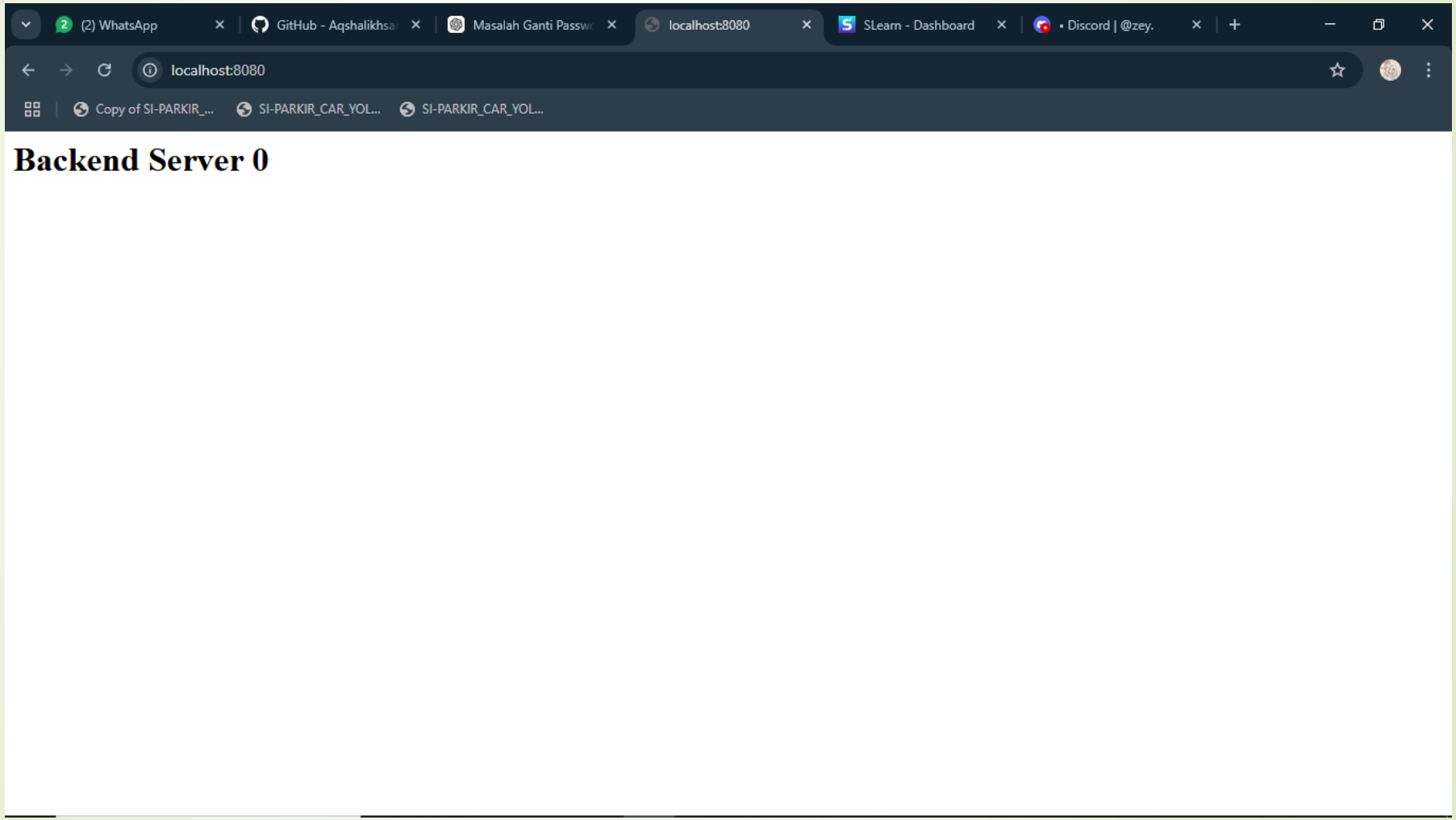
Terraform.tfState



Variable.tf


File Edit Selection View Go Run
terra_lab
EXPLORER
TERRA_LAB
.terraform
modules
providers
lb
nginx.conf
modules
backend
main.tf
variables.tf
loadbalancer
main.tf
variables.tf
.terraform.lock.hcl
main.tf
outputs.tf
terraform.tfstate
variables.tf
OUTLINE
TIMELINE
variables.tf
variables.tf
variable "server_count" {
default = 5
}
Ln 1, Col 1
Spaces: 2
UTF-8
CRLF
Plain Text
Finish Setup
Prettier

Output


(2) WhatsApp
GitHub - Aqshalikhsa
Masalah Ganti Passw
localhost:8080
SLearn - Dashboard
• Discord | @zey.
localhost:8080
Copy of SI-PARKIR_...
SI-PARKIR_CAR_YOL...
SI-PARKIR_CAR_YOL...
Backend Server 0